DBT PORTFOLIO PROJECT

LAYER: TESTING / VALIDATION

DBT Source Validation

Testing | Source Validation | Development | Snowflake | Data Quality

Project Summary

A development-time validation query that performs a full SELECT against the raw healthcare claims source table. Used to confirm that the Snowflake connection is live, all 14 columns are accessible and populated, and row counts are as expected before running any model builds.

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1 PROJECT OVERVIEW

Model Name:	test_source_validation (analysis or ad-hoc)
Layer:	Testing / Validation
Purpose:	Confirm source connectivity and column availability pre-build
Runs Against:	{{ source('healthcare', 'claims') }} -- raw source, not staging
Audience:	DBT developers, Data Engineers during development and debugging

2 BUSINESS CONTEXT

Before running a full dbt build, it is best practice to validate that the upstream source is healthy and returning expected data. This query serves as a quick sanity check -- confirming column names match the sources.yml definition, data is populated, and no upstream schema changes have broken the contract.

- Run this before dbt build to catch source connectivity issues early.
- Validates all 14 columns defined in sources.yml are physically present in the table.
- Useful after source schema migrations to confirm nothing broke silently.
- Can be extended with WHERE clauses to spot-check recent date ranges or specific entities.

3 TECHNICAL EXPLANATION

The query is intentionally minimal -- a SELECT * via the dbt {{ source() }} macro. This means it runs through dbt's connection profile (Snowflake warehouse, database, and schema) and will fail loudly if any part of the source declaration in sources.yml does not match the physical table.

- {{ source('healthcare', 'claims') }} resolves to healthcare_db.healthcare_schema.claims at compile time.
- SELECT * is used deliberately -- this is a diagnostic tool, not a production model.
- Should be placed in the analyses/ folder in the dbt project to avoid being compiled as a model.
- Results are inspected in the Snowflake query UI or dbt Cloud IDE preview pane.

4 FULL QUERY CODE

```
-- Source validation query
-- Purpose: confirm all columns are accessible before running dbt build
-- Place in: analyses/ folder (not compiled as a model)

select *
from {{ source('healthcare', 'claims') }}
```

5 VALIDATION CHECKLIST

Check	Expected Result	Status
Query returns rows	Row count > 0	Verify manually
period_date is DATE type	No string dates in output	Confirm cast
metric_value is numeric	No NULL-only column	Spot check
All 14 columns present	Column count = 14	Count columns
entity_id has no NULLs	0 NULL rows	Check WHERE entity_id IS NULL
is_fraud_flag is 0 or 1 only	No values outside {0,1}	Check DISTINCT